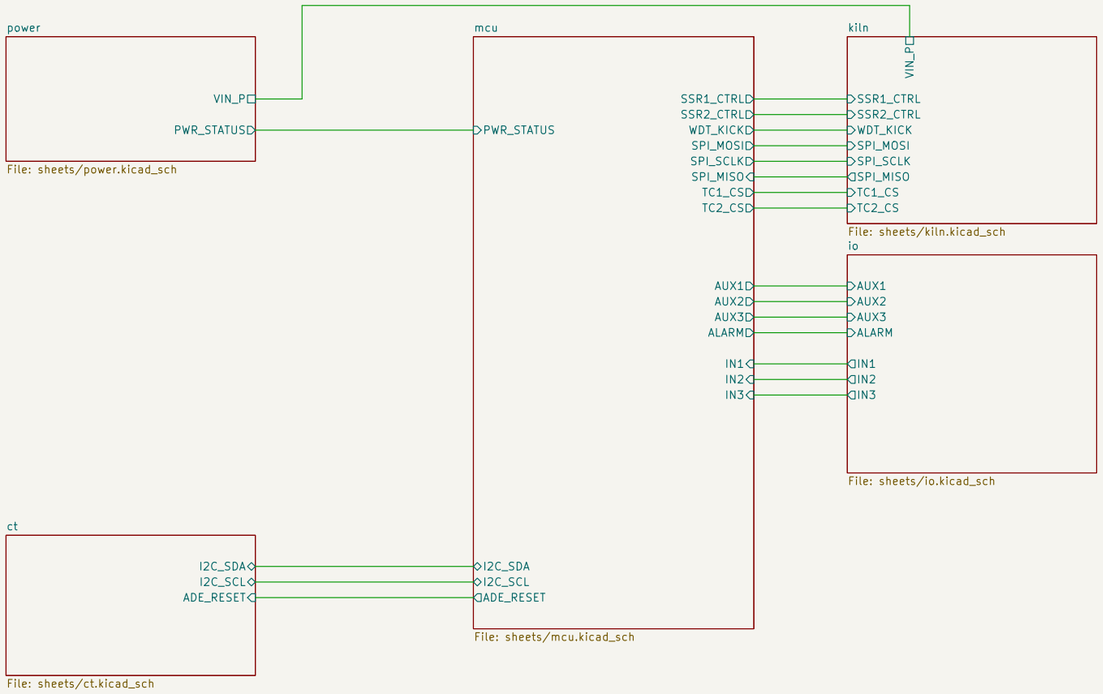



## NOTES — Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = VIN_P (24 V, fused and TVS-clamped, ALWAYS LIVE), pin2 = the switched low side. The watchdog gates the shared return (Q7), not the supply, so a tripped watchdog leaves 24 V on pin 1 and opens pin 2. Boot-safe: R7/R20 hold each gate down while the optos are dark. The optos are NOT an isolation barrier — both sides share GND. Rev B's opto-ISOLATED design was reverted and stays reverted; U8/U9 are here to keep a 24 V gate fault off the GPIO. Per channel <=200 mA: F1 holds at 750 mA and U11 takes ~250.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- — both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required; two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted; inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND — dry contact each channel to GND.

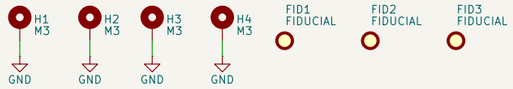
Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RST 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO 10=T_CLK 11=T_CS 12=T_DIN 13=T_DO 14=T_IRQ. PIN 1 IS +5V, NOT 3V3 — the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R39-R43 damping). LCD_RST is GPIO46, a boot-strapping pin: the MSP4021 has NO pull-up on its RESET line (QDtech schematic, 2019-09-27), so download mode is unaffected — check that before substituting another module.

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N — one CT clamp per SSR zone.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted — do not wire to AC mains. Y1 is a packaged 3.579545 MHz oscillator (no load caps; C25 is its bypass) — see DESIGN-NOTES.md, Y1. ADE7953 I2C address 0x38 would collide with a PCF8574A added on J7/J14.

Design reasoning per block: hardware/kicad/DESIGN-NOTES.md.
Designators C8, C9, C26, D1, D2 and Q1 are retired, not missing: they are kept free so the notes and reviews still read.

## MOUNTING / FIDUCIALS
## FID1-3 are bare-copper pick-and-place targets, on no net

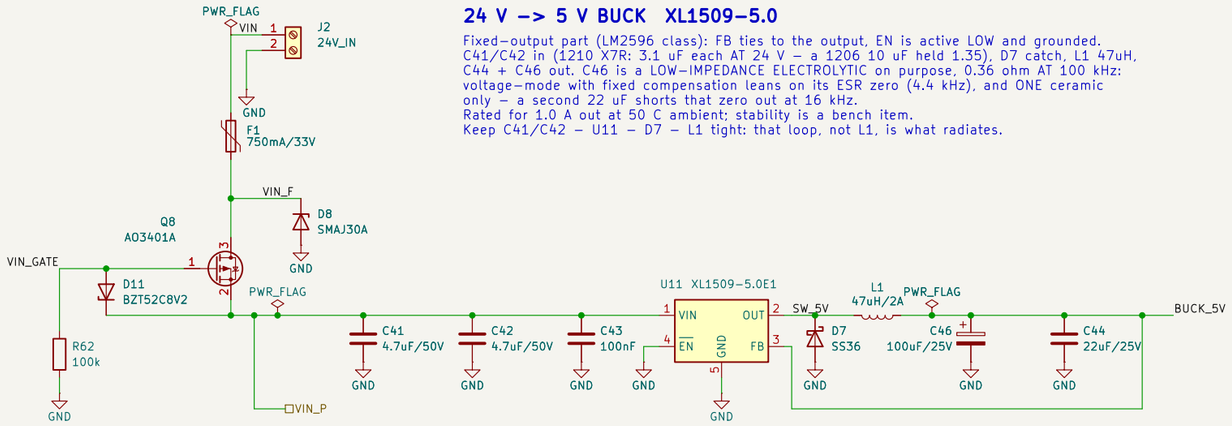



4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
**Bisque project**

| | | |
|---|---|---|
| Sheet: / | | |
| File: bisque-controller.kicad_sch | | |
| ***Title: Bisque Kiln Controller*** | | |
| Size: A3 | Date: 2026-10-22 | **Rev: B.5** |
| KiCad E.D.A. 10.0.7 | | Id: 1/6 |

## POWER IN  24 VDC terminal

F1+D8 clamp-and-blow at the entry. Q8 blocks reverse polarity as a P-FET, not a diode: its body diode conducts on correct polarity and the gate then pulls itself on through R62, clamped by D11 because 24 V is twice the AO3401A's V_GS rating.




## 24 V -> 5 V BUCK  XL1509-5.0

Fixed-output part (LM2596 class): FB ties to the output, EN is active LOW and grounded. C41/C42 in (1210 X7R: 3.1 uF each AT 24 V — a 1206 10 uF held 1.35), D7 catch, L1 47uH, C44 + C46 out. C46 is a LOW-IMPEDANCE ELECTROLYTIC on purpose, 0.36 ohm AT 100 kHz: voltage-mode with fixed compensation leans on its ESR zero (4.4 kHz), and ONE ceramic only — a second 22 uF shorts that zero out at 16 kHz.
Rated for 1.0 A out at 50 C ambient; stability is a bench item.
Keep C41/C42 — U11 — D7 — L1 tight: that loop, not L1, is what radiates.

## 5 V SOURCE SELECT  TPS2116, buck has priority




U12 selects between BUCK_5V and USB VBUS to make +5V, giving the buck PRIORITY via R63/R64 rather than letting the higher voltage win — a 5.25 V host would otherwise source the SSR terminals and the relay bank. U12 also blocks USB back-feeding the unpowered buck. C47/C48 are its input caps, one per channel, at the pins. C50 is the VBUS bulk by J1: U4 steers USB ESD into VBUS, and 3.2 uF at 5 V holds an 8 kV contact discharge to ~0.3 V on a pin rated 6 V.

4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953

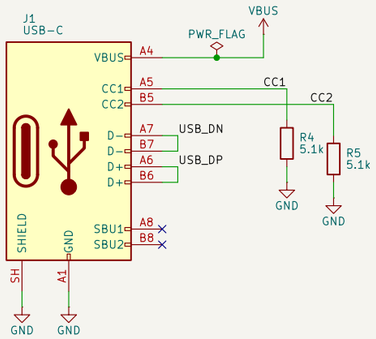
**Bisque project**

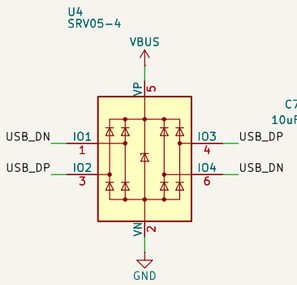
Sheet: /power/
File: power.kicad_sch

***Title: Bisque Kiln Controller — Power input and 24 V -> 5 V buck***

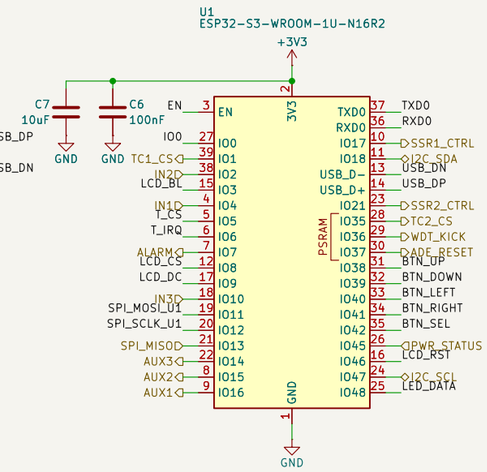
| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-22 | **Rev: B.5** |
| KiCad E.D.A. 10.0.7 | | Id: 2/6 |

## USB-C
native USB flashing + ESD

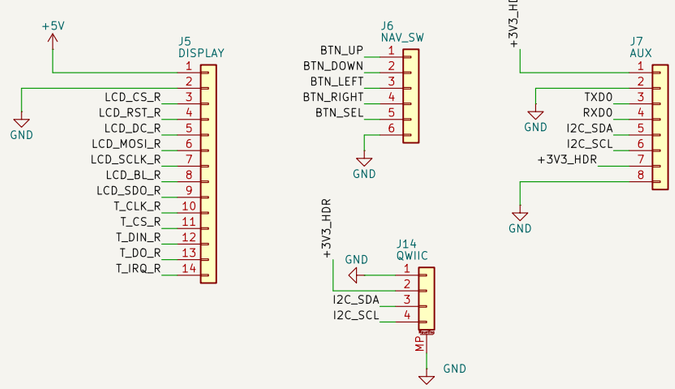


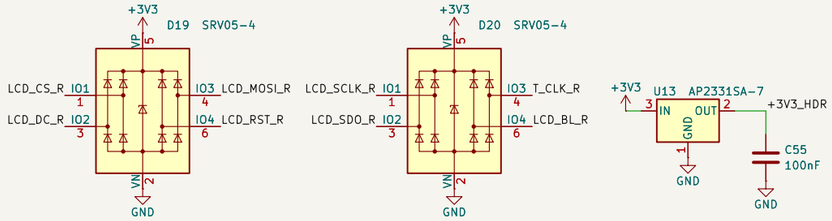

## ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults
R2 (quad PSRAM) or N16 modules only:
GPIO35/36 are PSRAM DQ4/DQ5 on R8



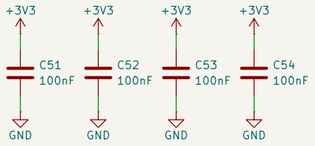

## HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C, J14 Qwiic I2C



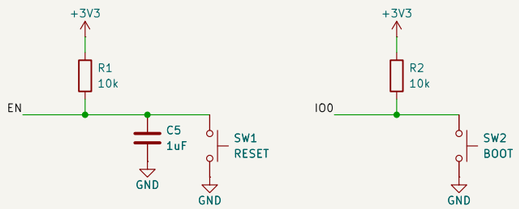

## DISPLAY LOOM ESD + HEADER 3V3 LIMIT  J5, J7/J14




## SPI RETURN-PATH CAPS (PCB review 2026-10-05 item 5)
C51-C54 sit on the board at the SPI bus's F.Cu/B.Cu transitions, not at U1: a signal that changes outer layer changes reference plane (In1 GND to In2 +3V3), and the return current crosses between the planes through the nearest 3V3-GND capacitor. One 0603 per via cluster, each with its own via to each plane.



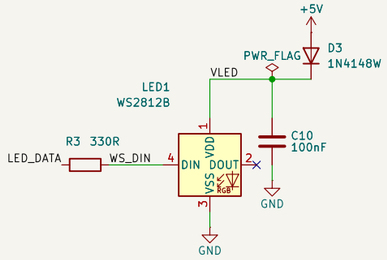

## RESET / BOOT



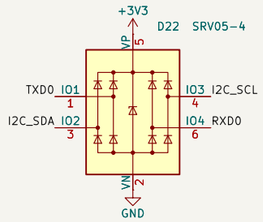

## +5V BULK

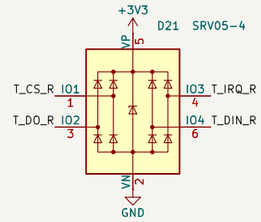



## WS2812B STATUS LED
VDD dropped ~4.6V for 3.3V data margin



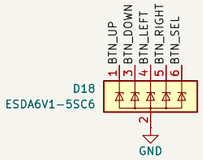

## J7/J14 + J5 TOUCH ESD



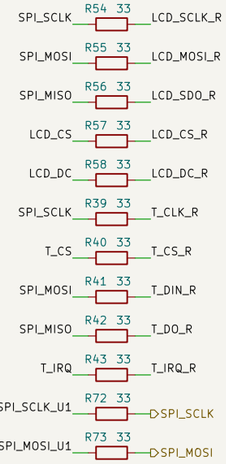




## LCD RESET / BACKLIGHT SERIES R + NAV ESD  J5, J6

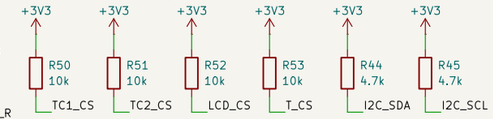



## SPI SERIES DAMPING




## PULL-UPS  chip selects idle high, I2C




## SIGNAL TEST POINTS
SPI, I2C, the two SSR control lines and a GND (TP3)



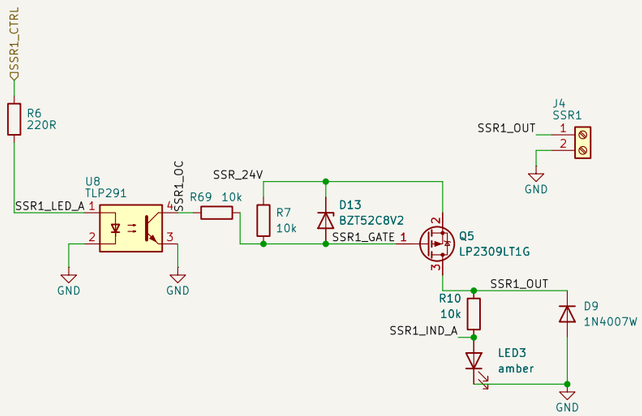

R54-R58 damp the four 40 MHz display lines and R39-R43 the display module's XPT2046 touch lines; both loom to J5. R72/R73 series-terminate SCLK and MOSI at U1's own pins, ahead of every branch: SPI_SCLK_U1/SPI_MOSI_U1 are the pin-side stubs.
R56 is also the SDO hedge - lift it if the panel does not tri-state J5.9 off the shared SPI_MISO. R50-R53 hold all four chip selects deasserted from power-on, before firmware configures any GPIO. J14 Qwiic and J7 5-8 (0.1 in) share I2C, pulled up by R44/R45.

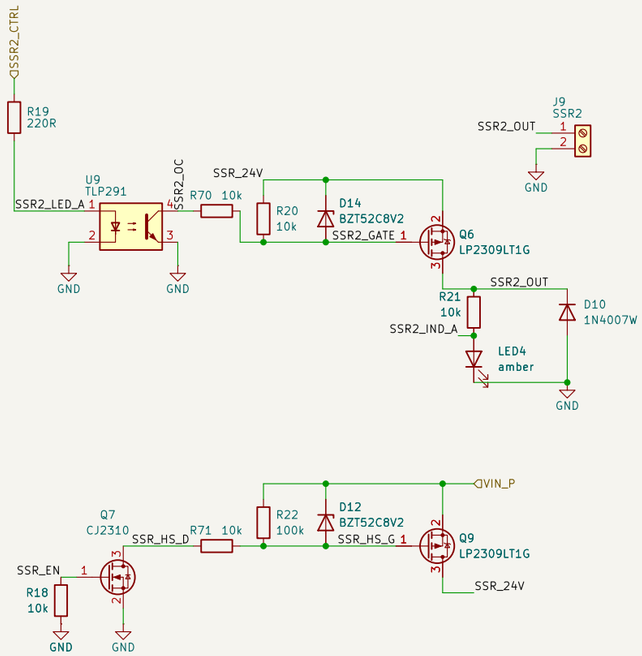
4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
**Bisque project**

| | | |
|---|---|---|
| Sheet: /mcu/ | | |
| File: mcu.kicad_sch | | |
| ***Title: Bisque Kiln Controller - ESP32-S3, USB, headers, status LED*** | | |
| Size: A3 | Date: 2026-10-22 | **Rev: B.5** |
| KiCad E.D.A. 10.0.7 | | Id: 3/6 |

## SSR DRIVE x2  high side, ground-referenced (rev B.4)

Each terminal's + is switched by a 60 V P-channel LP2309LT1G (Q5/Q6)
off SSR_24V; the - pin is board GND, so an SSR lead chafed onto the
earthed enclosure is a non-event, not a bypass. The GPIO drives an opto
LED, never a gate: at 24 V a gate-to-drain failure would put 240 mA
into the pin. The opto pulls the gate down through 10k (R69/R70); the
8.2 V zener (D13/D14) sets V_SG from the rail, not from CTR, and R7/R20
hold the gate at the source against the opto's dark current. D9/D10
freewheel a contactor coil; LED3/LED4 read the terminal itself.
NOT an isolation barrier - see DESIGN-NOTES.md (SSR block).



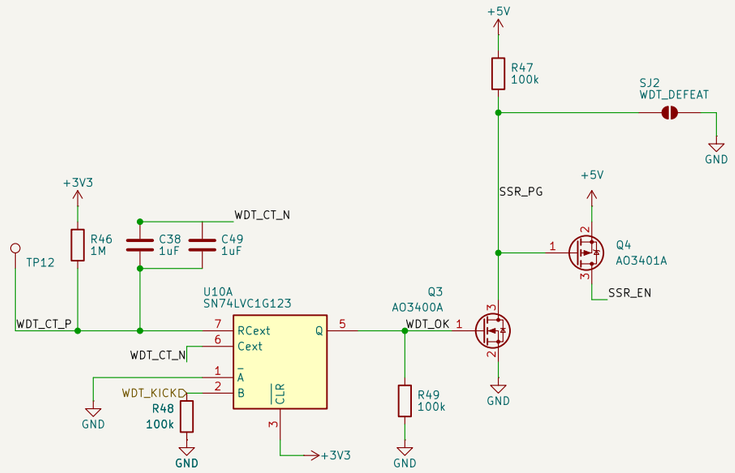



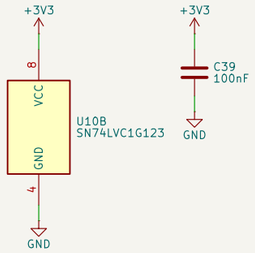




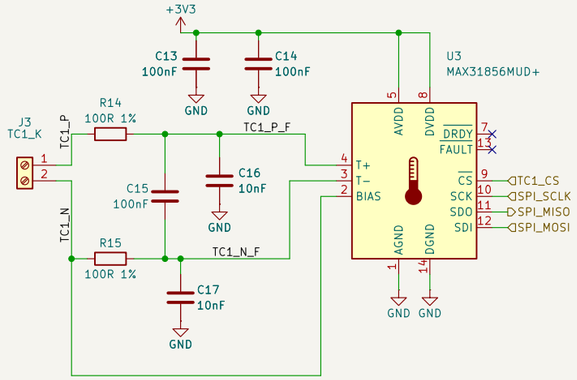
## HARDWARE WATCHDOG

U10 retriggerable one-shot: every rising edge on WDT_KICK restarts a
~2.2 s window, so Q stays high only while firmware keeps kicking.
Q holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which
supplies SSR_EN (+5 V). SSR_EN enhances Q7, which pulls the gate of the
24 V high-side switch Q9 down through R71 (D12 clamps V_SG to 8.2 V);
Q9 feeds SSR_24V to both channels. Window expired: SSR_EN falls, R22
pulls Q9's gate to its source, SSR_24V is dead and both terminals are
off whatever the channel switches are doing. R46/C38/C49 set the window.
R48 holds B low when the MCU is high-Z. R47 is the fail-safe pull-up.
SJ2 = bring-up defeat, REMOVE for service.

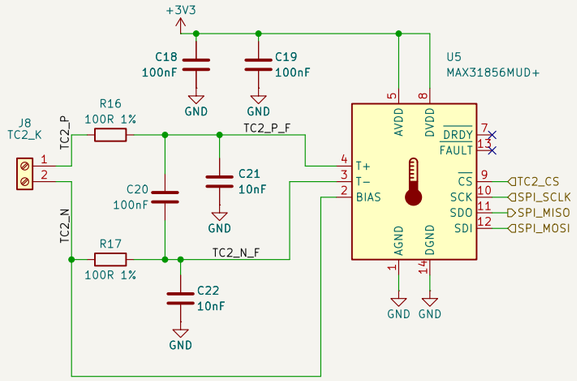



## THERMOCOUPLE 1 (control)  MAX31856
## T- floats to J3, biased via BIAS




## THERMOCOUPLE 2 (load)  MAX31856
## T- floats to J8, biased via BIAS




4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953

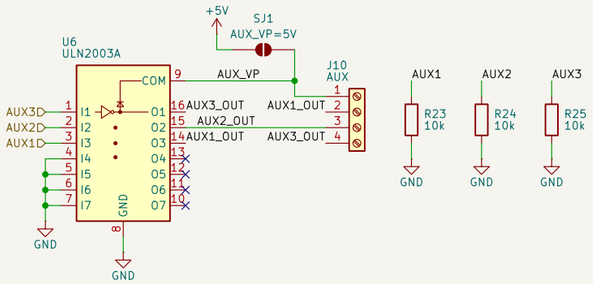
**Bisque project**

Sheet: /kiln/
File: kiln.kicad_sch

***Title: Bisque Kiln Controller — Thermocouples, SSRs, watchdog***

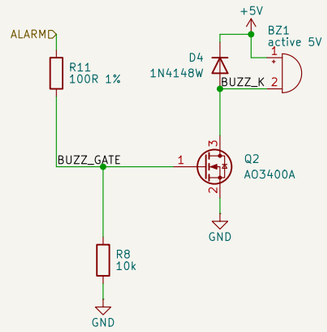
| Size: A3 | Date: 2026-10-22 | **Rev: B.5** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/6 |

## AUX OUTPUT BANK
## ULN2003 vent/purge/spare; COM->AUX_VP, SJ1 links to +5V

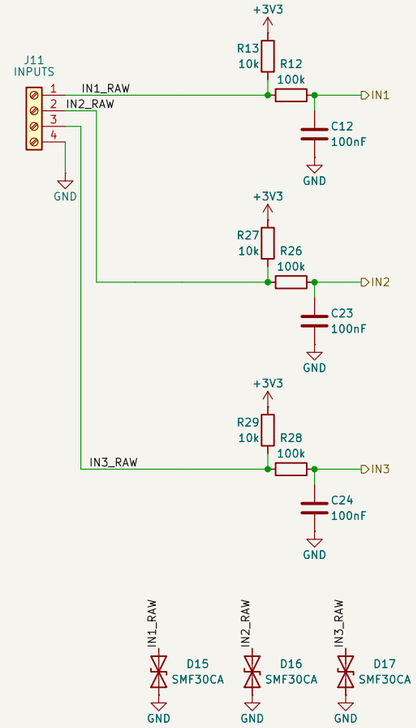



U6 channels run AUX3/AUX2/AUX1 on I1/I2/I3 (reversed against J10)
for routing; O1 and O3 reach J10 by label, O2 by wire.
R23–R25 hold each input low while the MCU pin is high-Z.

## ALARM BUZZER




## PROTECTED INPUTS x3  lid / gas-flow / spare
## 10k pull-up + SMF30CA at the terminal, 100k + 100nF to the pin



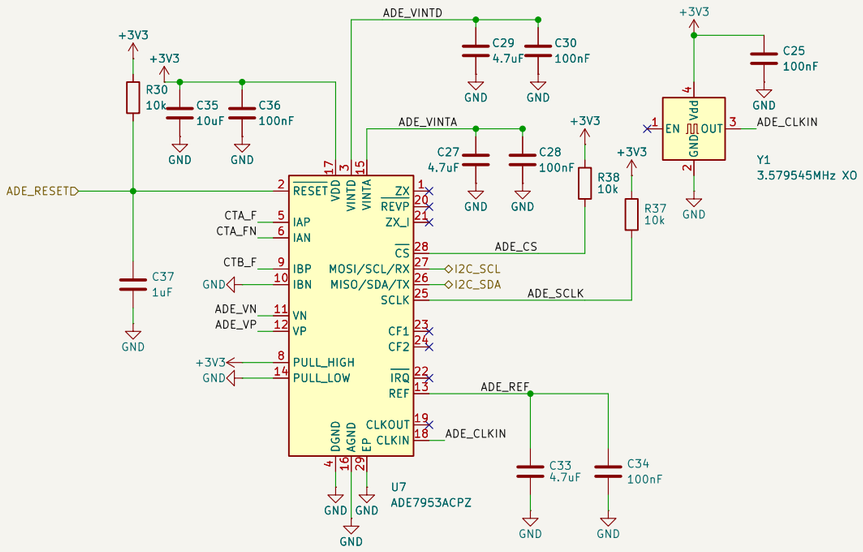

4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
**Bisque project**

Sheet: /io/
File: io.kicad_sch

***Title: Bisque Kiln Controller — Aux outputs, buzzer, protected inputs***

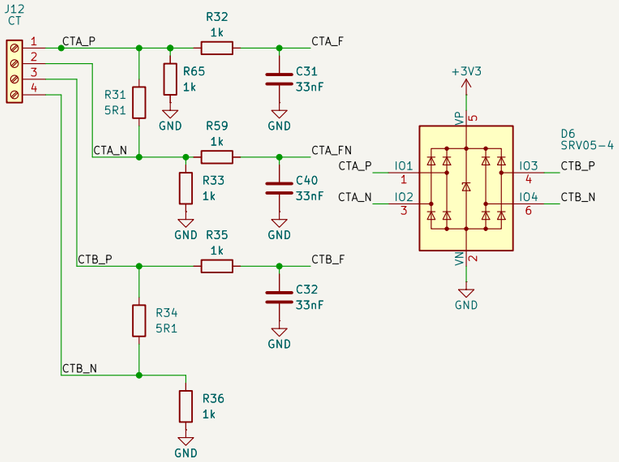
| Size: A3 | Date: 2026-10-22 | **Rev: B.5** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 5/6 |

## CT CURRENT SENSING  ADE7953, I2C, current-only
## no mains — VP/VN held at GND by R60/R61













4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
**Bisque project**

| | | |
|---|---|---|
| Sheet: /ct/ | | |
| File: ct.kicad_sch | | |
| ***Title: Bisque Kiln Controller — CT current sensing (ADE7953)*** | | |
| Size: A3 | Date: 2026-10-22 | **Rev: B.5** |
| KiCad E.D.A. 10.0.7 | | Id: 6/6 |